POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Fall

Year : 2013
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

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1. a) Write the buffer mechanism of basic buffer solution giving suitable example. Calculate the pH of the solution formed by mixing 500 ml of 0.2 M acetic acid to 500 ml of 0.4 m sodium acetate. Also calculate the pH of resulting solution when 1 ml of 1M HCI is added to the above solution. pKa for acetic acid is 4.74. 8

b) How is single electrode potential of Cu electrode measured experimentally? From the given electrode potential values answer the following: 7

i. Write the electrode reactions.

ii. Write the cell reaction.

iii. Calculate the e.m.f. of the cell at 27^0c when both the electrodes are coupled together

$E^0 Mg/Mg^{++} = +2.370v.$

$E^0 Fe/Fe^{++} = +0.440v$

$R = 8.314$ J mol^{-1} K^{-1}

$F = 96500C$

$[Mg^{++}] = 0.1M$, $[Fe^{++}] = 0.01M$

2. a) Give reasons: 8

i. Ionisation energy of nitrogen is greater than oxygen.

ii. Electron affinity of chlorine is higher than fluorine.

iii. Electro negativity of 'Ga" is higher than 'Al'.

iv. Atomic radii go on increasing while moving top to bottom in a group.

b) Explain why zinc sulphate salt is colorless whereas copper sulphate salt is colorful in nature. Why are transition elements mostly 7

paramagnetic? Explain.

3. a) Differentiate between enantiomers and diasteriomers with examples. What are carbocations? 3° carbocations are more stable than 2° carbocations why? 7

b) What are elimination reactions? Write the mechanism of E_2 reaction giving suitable example. 8

4. a) Write the mechanism of addition polymerization with example. Give the preparation, properties and uses of nylon 66. 7

b) Show your acquaintance on: 8

i. Vulcanization of rubber

ii. Nylon 6,6.

5. a) Write the procedure involved in TLC. How many NMR signals would you expect in following compounds: 8

i. $CH_3 - O - CH_3$

ii. CH_3CH_2OH

b) How cement can be manufactured? Write the stepwise chemical reaction involved during setting of cement. 7

6. a) Define water pollution. Mention and explain the causes and effects of water pollution. What can be done to control it? 7

b) What are green house gases? How are these gases responsible for global warming? Explain the effects of global warming and how can it be controlled? 8

7. Write short notes on: (Any two) 2×5

a) Electrochemical series

b) Determination of hardness of water.

c) Bakelite.